

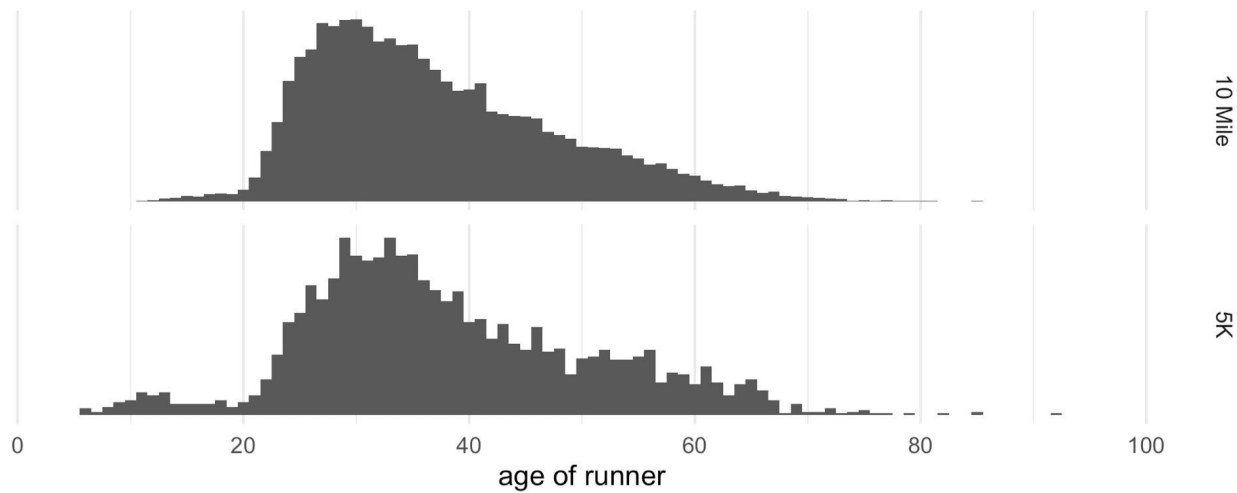
Cherry Blossom Population Questions (Ages)

MATH 213

Group 1

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were 1 year younger than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: $10.7/\sqrt{50} = 1.513$

SE for 5K: $12.7/\sqrt{50} = 1.796$

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: -1

Shape: Normal

Spread: $\sqrt{(1.513^2 + 1.796^2)} = 2.348$

$$2.348 * 2 = 4.696$$

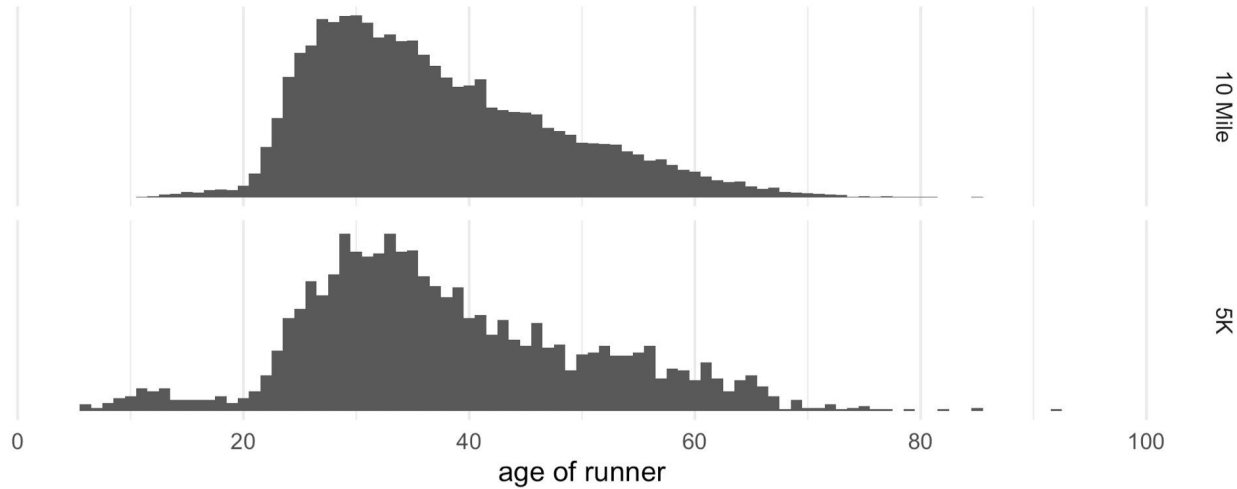
$$q_{025} = -1 - 4.696 = -5.696$$

$$q_{975} = -1 + 4.696 = 3.696$$

Group 2

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were [1] years [younger] than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: 1.513

SE for 5K: 1.796

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: ~37.5

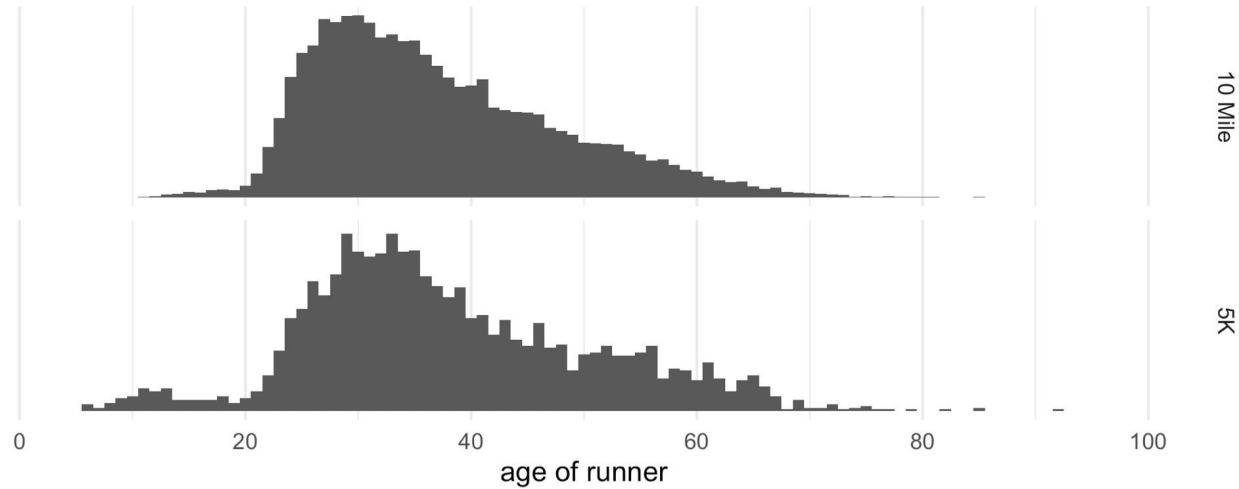
Shape: Bell-shaped

Spread: ~11.7

Group 3

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were 1 years **younger** than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: $10.7/\sqrt{50} = 1.51$

SE for 5K: $12.7/\sqrt{50} = 1.80$

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: The center will be the difference in the population means which is at the -1 year mark.

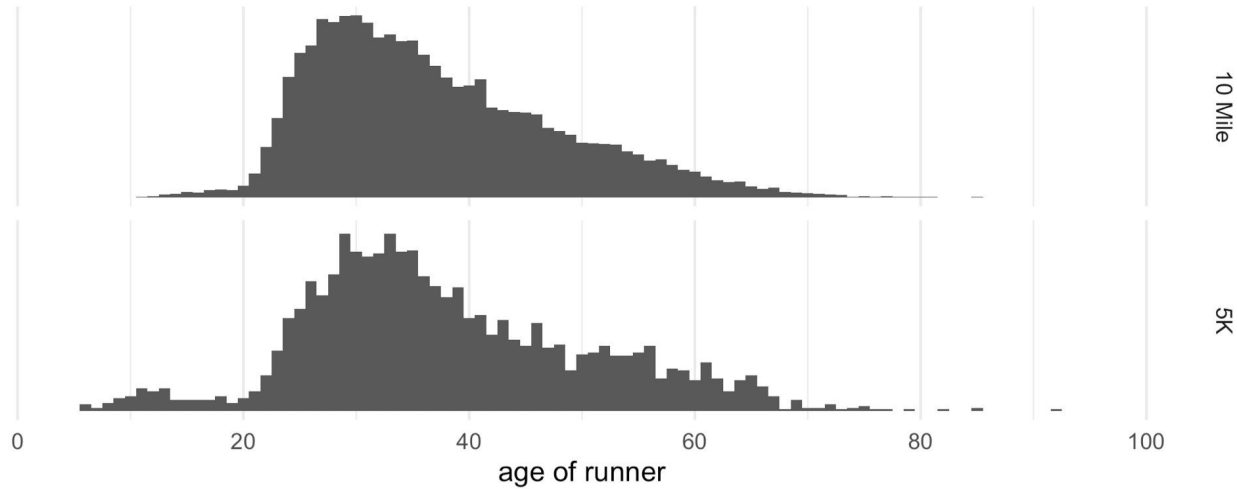
Shape: Should be normal, bell-shaped.

Spread: The standard error for the difference in the sample mean is $\text{square root}(1.51)^2 + (1.80)^2 = 2.35$

Group 4

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were **1** year **younger** than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: $SD/\sqrt{n} = 10.7/\sqrt{50} = \mathbf{1.5}$

SE for 5K: $12.7/\sqrt{50} = \mathbf{1.8}$

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: Center should be roughly at the 10mi - 5k difference in means for the real world race data. **-1**

Shape: Shape should be mostly normal with a skew to the right as there are some individuals who continue to race well into their older ages and there's no telling how many or how few of them may be picked in the simulations.

Spread: $SE_{diff} = \sqrt{SE_{10}^2 + SE_{5k}^2} = \sqrt{1.5^2 + 1.8^2} = 2.34$

$-1 - 2 * 2.34 = -5.68$

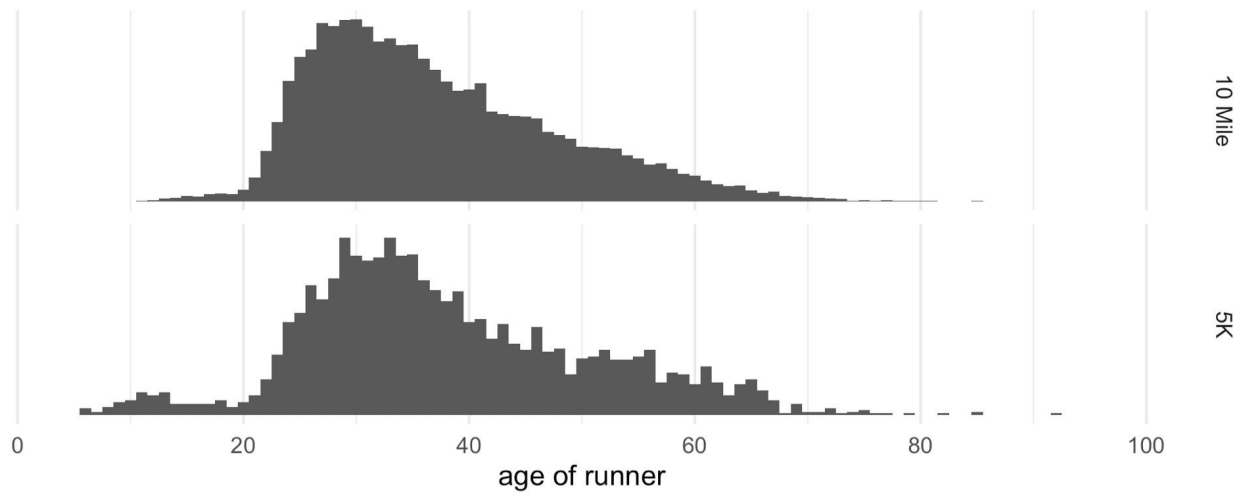
$-1 + 2 * 2.34 = 3.68$

95% confidence interval: (-5.68, 3.68)

Group 5

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were **one** year **younger** than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: $s/\sqrt{n} \rightarrow 10.7/\sqrt{50} = 1.513$

SE for 5K: $s/\sqrt{n} \rightarrow 12.7/\sqrt{50} = 1.796$

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: difference in the means

Shape: The shape may have skew due to outliers in age that are participating in the race.

Spread: $\text{sqroot}((1.513)^2 + (1.796)^2) = 2.348$

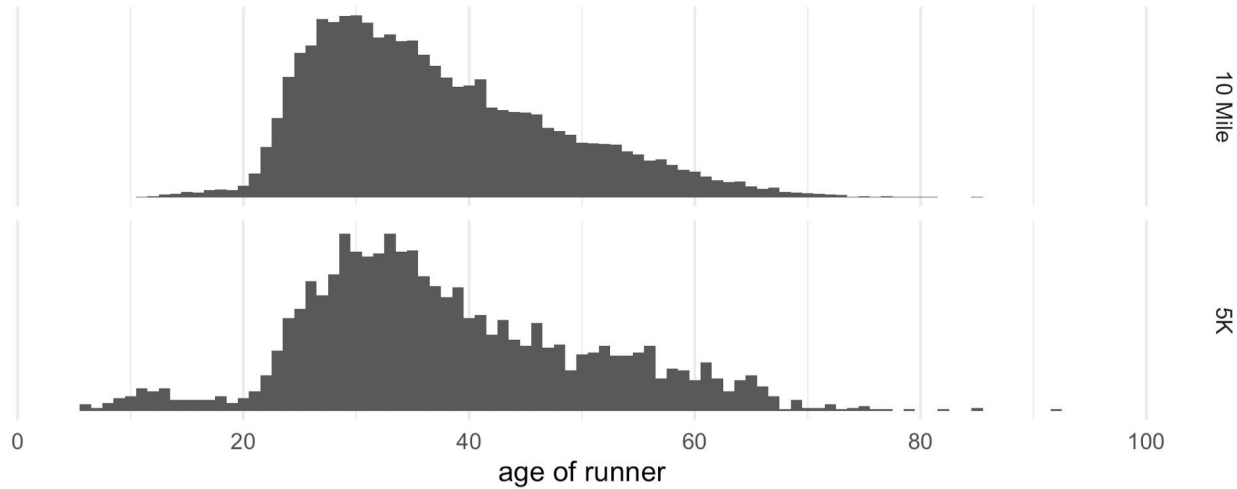
$-1 - 2(2.348) = -5.696$

$-1 + 2(2.348) = 3.696$

Group 6

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were [1] years [younger] than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: 1.51

SE for 5K: 1.80

We will look at the sampling distribution for the **difference in sample means** (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: The population's difference in means (-1)

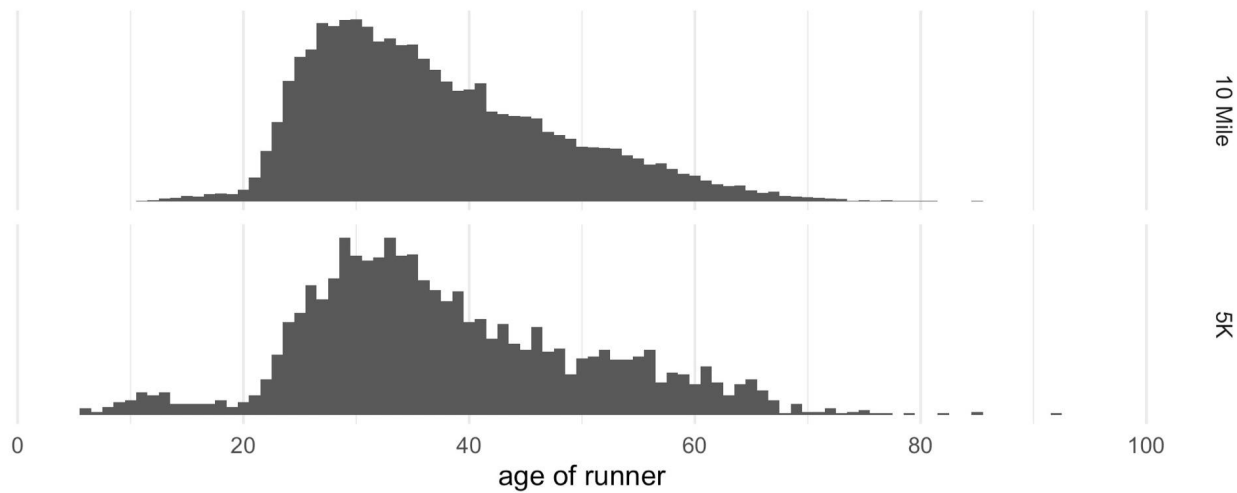
Shape: Bell shape

Spread:

Group 7

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were [1] years [younger] than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: 1.5

SE for 5K: 1.8

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: Around 0

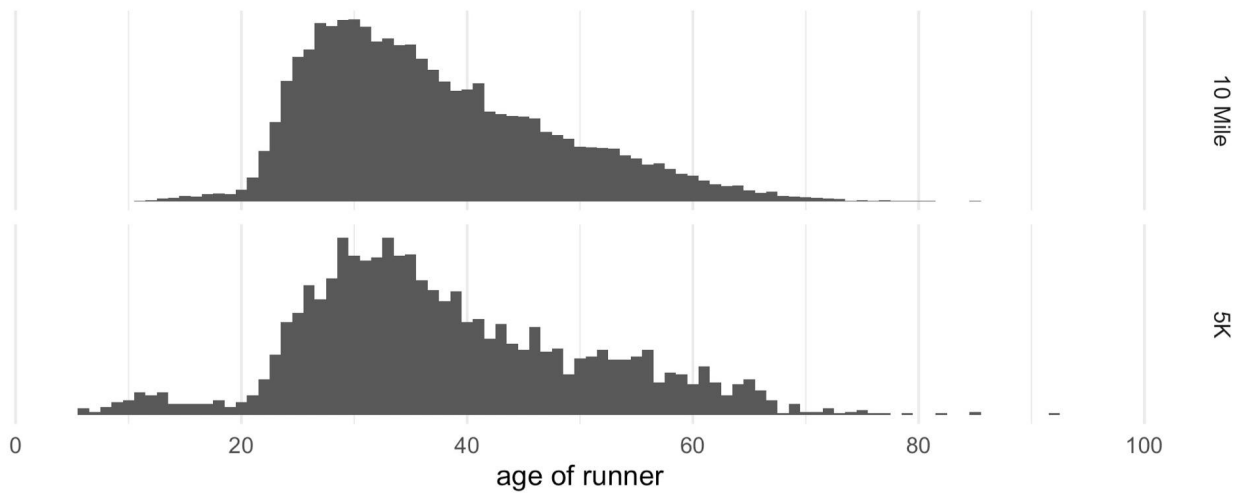
Shape: Bell shaped

Spread: Not a lot of outliers

Group 8

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were 1 years younger than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?
SE =

SE for 10 mile: $10.7 / \sqrt{50} = 1.513$

SE for 5K: $12.7 / \sqrt{50} = 1.796$

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: around -1

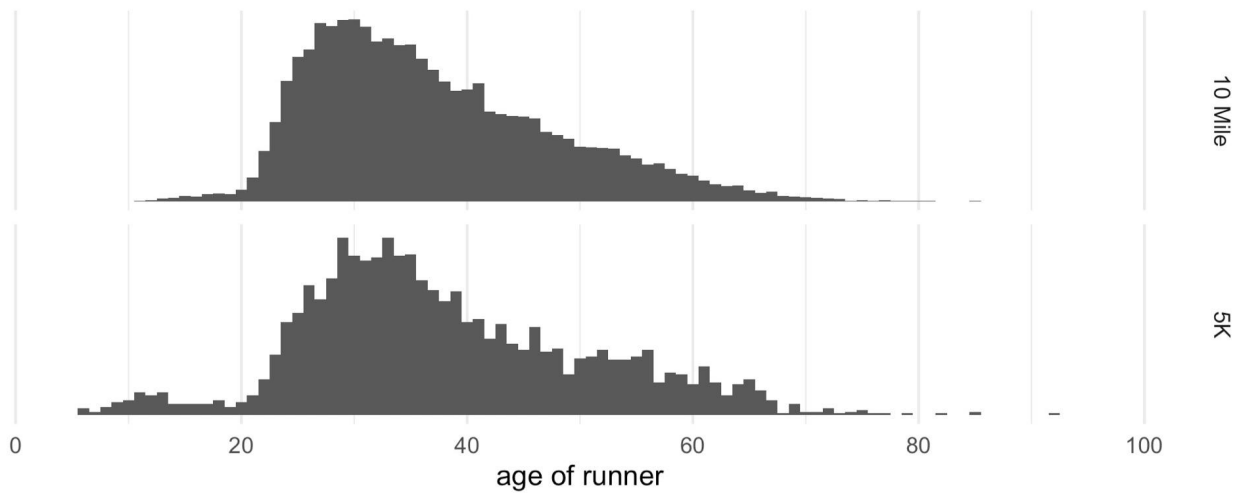
Shape: bell shaped

Spread:

Group 9

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were 1 years younger than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: $10.7 / \sqrt{50} = 1.513$

SE for 5K: $12.7 / \sqrt{50} = 1.796$

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: -1

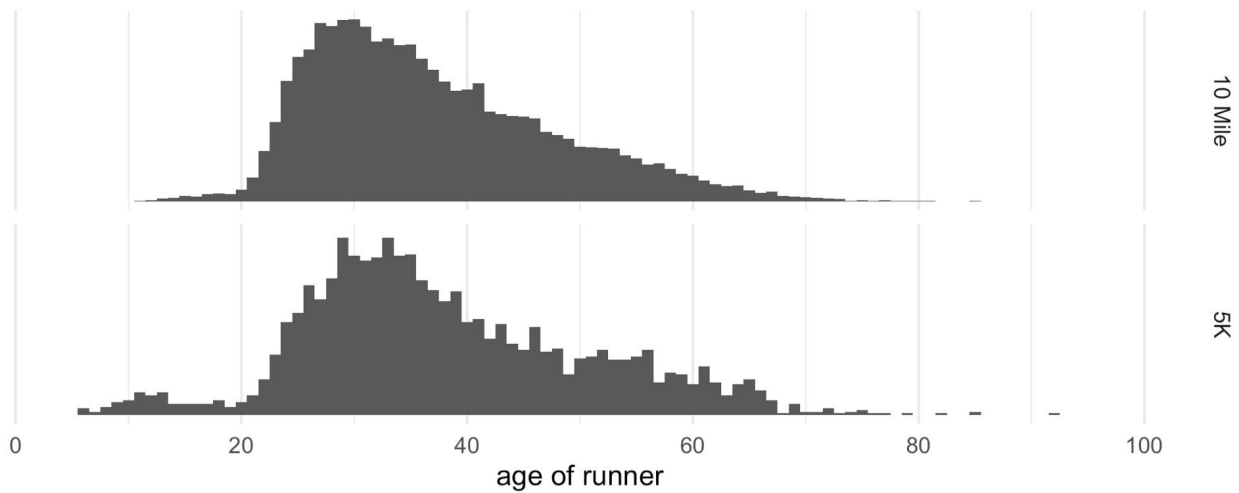
Shape: bell shaped

Spread:

Group 10

Population

Runners in 2017 Cherry Blossom running festival



Event	Mean age	Standard dev. age
10 mile	37.0	10.7
5K	38.0	12.7

Complete the following sentence describing the difference-in-means parameter:

On average, people in the 10-Mile race were 1 years younger than people in the 5K race.

We will use this population to study the “difference in means” statistic. We will look at pairs of samples: one sample of 50 people from the 10-mile group and one sample of 50 people from the 5K group.

We will make a sampling distribution by calculating the difference in mean ages (10 mile – 5K) for thousands of such pairs of samples.

Imagine taking a random 50-person sample from the 5K runners and a random 50-person sample from the 10-mile runners.

According to our standard-error formulas, what are the standard errors for the sample mean ages?

SE for 10 mile: $se = 10.7/7.071 = 1.513$

SE for 5K: $se = 12.7/7.071 = 1.796$

We will look at the sampling distribution for the difference in sample means (10-mile – 5K) for this kind of sampling. Based on your experience with sampling distributions, what do you expect to see?

Center: The center will be -1 μ of 10k - μ of 5k = -1

Shape: The shape is going to be bell shaped in the center of the graph

Spread: The spread will be around 9